Harsha Vardhan Yellela

United States | +1-248-497-9965 | harsha.yellela@gmail.com | har5ha.in | LinkedIn | GitHub

SUMMARY

DevOps Engineer with hands-on experience in **Kubernetes orchestration**, **CI/CD pipeline development**, and **Infrastructure as Code (Terraform, CloudFormation)**. Proficient in **Docker containerization**, **AWS cloud services (EKS, ECS, Lambda)**, and **monitoring solutions (Prometheus, Grafana)**. Built production systems with **autoscaling (HPA)**, **circuit breaker patterns**, and **8-microservice architectures**. Strong foundation in **automation**, **GitOps workflows**, and **reliability engineering**.

EXPERIENCE

Student Consultant – ML Infrastructure **Feb 2026 – Present**
Open Avenues (Build Fellowship) *Remote*

- Training a GPT language model on **dual NVIDIA RTX 3090 GPUs** using torch.nn.DataParallel; managing GPU environment, checkpoint persistence, and training observability (loss curves, step logging across 25K steps).
- Deploying trained model weights to **HuggingFace Hub** and serving as a **Gradio web app on HuggingFace Spaces** — public zero-install inference endpoint with runtime weight download.
- Engineered a **reproducible end-to-end ML pipeline**: web scraping → preprocessing → tokenizer training → model training → deployment, with configurable hyperparameters and switchable tokenizer backends.

Graduate Research Assistant – Agentic AI **Jan 2025 – Dec 2025**
Lawrence Technological University *Southfield, MI*

- Deployed persistent **backend services** on **AWS Fargate** and **Amazon EKS**, managing container orchestration and service mesh configurations.
- Configured **OpenSearch Serverless** clusters for semantic search, implementing **auto-scaling policies** and **access control**.
- Built **CI/CD pipelines** for multi-agent systems using **GitHub Actions**, achieving **70% reduction in deployment time**.

Infor India Pvt. Ltd. **Apr 2022 – Dec 2023**
LN Technical Consultant *Hyderabad, India*

- Integrated **AWS S3, Lambda, and API Gateway** with **Infor ION** process flows for asynchronous file transfer and event-driven automation.
- **Containerized** business logic services using **Docker** and simulated **Kubernetes-like orchestration** with enterprise systems.
- Developed **automation scripts** for global clients (**Ferrari, Boeing, Triumph**), reducing **batch processing delays by ~40%**.

Junior Java Trainee (PEP Program) **Sep 2020 – Jul 2021**
EPAM Systems *Hyderabad, India*

- Completed structured **Java Core** training with exposure to **CI/CD concepts** — build pipelines, automated testing, and deployment basics — forming the foundation for later container-based and infrastructure automation work.
- Operated in **Agile/Scrum** sprints with **Git-based version control**; debugged service behavior in **Linux environments** using CLI tooling and log-based analysis.

TECHNICAL SKILLS

- **Container Orchestration:** Kubernetes (EKS), Docker, Docker Compose, Helm | HPA, PVC, Deployments, Services
- **CI/CD:** Jenkins (Multibranch Pipelines), GitHub Actions, GitOps | Automated Testing, Deployment Strategies
- **Infrastructure as Code:** Terraform, AWS CloudFormation, AWS CDK | State Management, Modular Configurations
- **Cloud (AWS):** EKS, ECS Fargate, Lambda, EC2, API Gateway, S3, DynamoDB, ECR, SQS, SNS, ALB
- **Monitoring & Observability:** Prometheus, Grafana, CloudWatch | Metrics, Alerting, Dashboards, Log Aggregation
- **Scripting & Languages:** Bash, Python, Go, YAML | Shell Scripting, Automation, Configuration Management
- **Reliability Patterns:** Circuit Breaker, Health Checks (Readiness/Liveness), Auto-scaling, Spot Instances
- **Practices:** GitOps, Agile/Scrum, Infrastructure Automation, Security Best Practices, Cost Optimization

PROJECTS

ML Sentiment Feedback Loop – Production MLOps Microservices **Dec 2025**
Terraform, GitHub Actions, AWS (ECS Fargate, SageMaker, ECR), Docker | GitHub

- Built **8-microservice architecture** with **independent scaling** on **AWS ECS Fargate**, implementing **API Gateway** for unified routing.
- Configured **GitHub Actions CI/CD** with **Terraform IaC** for automated infrastructure provisioning and container deployments to **ECR**.
- Designed **auto-deployment pipeline** (training → model registry → production) with **SNS notifications** and **SageMaker integration**.

Online Learning Portal – Microservices CI/CD Platform **Nov 2025**
Jenkins, Docker, Kubernetes, AWS (ECS Fargate, ECR, ALB) | GitHub

- Implemented **Jenkins Multibranch Pipeline** with **branch-per-service CI strategy**, enabling independent Docker builds for **6 microservices**.
- Created **unified CD pipeline** for **EKS deployment** with **LoadBalancer service** configuration and **automated Docker builds** with commit hash tagging.
- Designed **Kubernetes manifests** for all services with **webhook automation** for continuous integration triggers.

Telegram Toxicity Moderator – Production K8s ML Platform **Oct 2025**
Kubernetes (EKS), Prometheus, Grafana, FastAPI, HPA, ALB | GitHub

- Deployed **ML inference service** on **AWS EKS** with **Horizontal Pod Autoscaler (HPA)** achieving **500 RPS average, 1500 RPS peak** capacity.
- Implemented **circuit breaker pattern** (5 failures, 30s recovery window) with **readiness/liveness probes** for production reliability.
- Created **Prometheus metrics** and **Grafana dashboards** for real-time monitoring; reduced costs **70%** with **spot instances** and aggressive scale-down.

Car Dealer App – Full-Stack Kubernetes Application **Oct 2025**
Kubernetes, Docker, PostgreSQL, Flask, Nginx | GitHub

- Configured **3-tier Kubernetes architecture** with **Deployments, Services, PersistentVolumeClaims** and **2 replicas** for high availability.
- Implemented **readiness/liveness probes** for health monitoring and **LoadBalancer service** for external traffic routing.
- Wrote comprehensive **shell scripts** for **deployment automation** including build, push, and rollout commands.

Job Portal REST API – Containerized Cloud Deployment **Oct 2025**
Docker, Docker Compose, AWS ECS/ECR, FastAPI, GitHub Actions | GitHub

- **Containerized** application using **Docker** and **Docker Compose** with **persistent named volumes** for data durability across restarts.
- Deployed to **AWS cloud infrastructure** using **ECR** for container registry and **ECS** for orchestration with end-to-end **CI/CD pipeline**.
- Built **production-ready API** with **25+ endpoints**, comprehensive **OpenAPI/Swagger documentation**, and **shell scripting automation**.

Lambda Microservices Platform – Serverless Infrastructure **Dec 2024**
AWS Lambda, Terraform, CloudFormation, DynamoDB, API Gateway, S3

- Developed and deployed **94 AWS Lambda functions** for complete SaaS platform using **Terraform** and **CloudFormation** templates.
- Built **automated deployment pipeline** with **Git change detection**, deploying only modified functions to reduce deployment time.
- Configured **API Gateway** with Lambda integrations, **DynamoDB streams**, and **S3 event triggers** for event-driven architecture.

LuffyGPT – GPT Transformer Language Model from Scratch **Jan 2026 – Mar 2026**
Python, PyTorch, SentencePiece, tiktoken, Gradio, HuggingFace Hub | GitHub | Demo

- Implemented a **GPT decoder-only transformer from scratch** in PyTorch (10.81M params) — multi-head causal self-attention, positional embeddings, FFN with 4× expansion, residual connections, pre-LayerNorm, dropout; followed the *Attention is All You Need* architecture.
- Trained a **custom BPE tokenizer** using SentencePiece on a ~3.2M character corpus (vocab 2000, 3.11× compression); benchmarked char-level, GPT-2 BPE, cl100k, and o200k strategies; trained on **dual NVIDIA RTX 3090 GPUs** (DataParallel), reaching **val loss ~1.18** at 25K steps.
- Deployed as a **Gradio app on HuggingFace Spaces** with model weights hosted on HuggingFace Hub; built a CLI tool supporting `--train`, `--eval`, `--interactive`, and `--custom-bpe` modes.

EDUCATION

Lawrence Technological University **Jan 2024 – Dec 2025**
*Master of Science in Computer Science · GPA: **3.6/4.0*** *Southfield, MI*

- **Relevant Coursework: Cloud Microservices**, **Machine Learning**, **Artificial Intelligence**, **Intelligent Robotics (ROS)**

Geethanjali College of Engineering & Technology **Aug 2018 – Aug 2022**
*Bachelor of Technology in Computer Science & Engineering · GPA: **7.5/10 (~3.0/4.0)*** *Hyderabad, Telangana*

- **Relevant Coursework: Software Engineering**, **Operating Systems**, **Computer Networks**, **Database Management**

ACHIEVEMENTS

- **Selected for Amazon Nova AI Challenge: Trusted AI Track** (2025)
- **Built production SaaS platform** with 94 Lambda functions and Terraform IaC serving real customers (2024)
- **Gold Medalist in Indian National Mathematical Olympiad (INMO)** (2012)